ALOK SAHOO

Software Engineer

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PROFESSIONAL SUMMARY

Final-year B.E. IT student with practical experience in backend and full-stack development, REST APIs, and databases. Skilled in TypeScript, Python, JavaScript, AI/ML, cybersecurity, and computer vision. Experienced in building and deploying real-world software through internships and technical projects.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, SQL

Frameworks: React.js, Next.js, Node.js, Express.js, Fastify, FastAPI

Databases: MongoDB, PostgreSQL, Elasticsearch

AI / Machine Learning: PyTorch, Hugging Face Transformers, OpenCV, Deep Learning, Computer Vision, XAI

Architecture & Systems: REST APIs, Microservices, WebSockets, JWT, OAuth

DevOps & Tools: Docker, Git, GitHub Actions, Kibana

AI: LLMs, Prompt Engineering

PROFESSIONAL EXPERIENCE

IMS Learning Resources Pvt Ltd

Feb 2026 -- Apr 2026

Software Engineer Intern — Remote

- Developed backend REST APIs using TypeScript and Fastify to handle structured routing and CRUD operations for assigned internal services.
- Integrated Elasticsearch for data indexing and search queries, evaluating retrieval workflows in local development environments.
- Managed version control with Git and GitHub, independently completing milestone implementation tickets and deploying builds to Vercel.

Sapphire Infocom Private Limited

Dec 2025 -- Mar 2026

Full-Stack Developer Intern — Remote

- Built the application backend from scratch using Node.js, Express.js, and MongoDB, designing database schemas and REST APIs for users, posts, and social interactions.
- Implemented secure JWT authentication, role-based admin access, custom middleware, input validation, and Cloudinary media upload integration.
- Wrote modular route handlers with centralized error handling and deployed the completed full-stack application on Vercel.

Techriciate (Early-Stage Startup) | techriciate.com

2026

Project-based Technical Contributor — Remote

- Supported an early-stage startup by independently developing its official portfolio website using Next.js, Tailwind CSS, and Framer Motion, including domain DNS, SSL, and business email setup.
- Contributed to a commercial web platform using Next.js, TypeScript, and MongoDB Atlas, implementing product catalogs, dynamic category filters, cart management, and WhatsApp order redirection.
- Built protected admin dashboard routes for product, category, and banner management with Cloudinary media integration, delivering demonstration milestone releases on Vercel.

SELECTED PROJECTS

TrustNet-AI | **Digital Trust & Deception Detection Platform**

Technologies: Python, FastAPI, React, PyTorch, ViT, OpenCV, Docker

- Developing an explainable digital-deception platform focused on image deepfake detection, architecting the end-to-end pipeline to identify synthetic faces and manipulated media.
- Integrated Hugging Face Vision Transformers (ViT) and EfficientNet spatial backbones with multi-layer forensics (2D FFT, Error Level Analysis, sensor noise) and Grad-CAM explainability across modular FastAPI microservices and React.

AURA | **Advanced URL Response Analyzer (SIH 2025 National Finalist)** GitHub

Technologies: React, Three.js, Kibana, Data Visualization, Python

- Built the analyst-facing frontend for a Smart India Hackathon 2025 National Grand Finale cybersecurity platform that inspects network traffic to distinguish attack attempts from confirmed breaches.
- Created interactive network and threat visualizations using React, Three.js, dynamic charts, and Kibana dashboards, curating the network traffic dataset required for SIH problem evaluation.

Reimbursement Automation System | **Institutional Workflow Platform** GitHub | Live Demo

Technologies: React.js, Node.js, Express.js, MongoDB, REST APIs

- Developed an institutional web platform to digitize academic expense submissions and streamline multi-tier approval workflows across 4 role-based portals (Student, Faculty, HOD, Principal).
- Generated PDF reports for submitted reimbursement records, contributed 20--30% of backend REST APIs and MongoDB schemas, and integrated full-stack workflows deployed for college departmental operations on Vercel.

ACHIEVEMENTS & HONORS

- **Smart India Hackathon 2025** | National Grand Finale Finalist, Ministry of Education / AICTE 2025
- **OpsStorm Hackathon** | 1st Prize Winner, Technical Systems & Automation Challenge 2026
- **IEEE Techithon ’26 (Hackdeck 2.0)** | 1st Rank, Atharva University Mumbai 2026
- **HackStreak 2.0** | First Place (Third Year Category), ITSA, A. P. Shah Institute of Technology 2026
- **DataWeb Hackathon** | 2nd Prize, Generative AI Domain, Intercollegiate 6-Hour Hackathon 2026

EDUCATION

A. P. Shah Institute of Technology Thane, Maharashtra, India

Bachelor of Engineering (B.E.), Information Technology Expected 2027

Academic Record: CGPA: 7.94 / 10.0 | Status: Final Year Undergraduate

Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks

Senior Capstone Project: TrustNet-AI | Explainable Digital-Deception Detection Platform

CERTIFICATIONS

- **Oracle Agentic AI Foundations (2026)** | Oracle University 2026
- **NPTEL DBMS (Elite) & Software Engineering** | NPTEL 2025
- **IBM Web Development Fundamentals** | IBM / IBM SkillsBuild 2025
- **Fortinet Network Security Associate Virtual Internship** | Fortinet 2025
- **AWS Data Engineering & Google Cloud Generative AI** | Virtual Internships 2025
- **Google Android Developer Virtual Internship** | Google Developers / EduSkills 2025
- **Java, Machine Learning & MERN Stack** | Success Classes of Engineering 2025